

When transporters own the road, how much can the network carry in a day?

The road-occupancy model and the network bound: thesis chapter 2 and the bound section of chapter 3

A capacity bound for shipyard road networks with shareable widths and asymmetric loaded and empty occupancy

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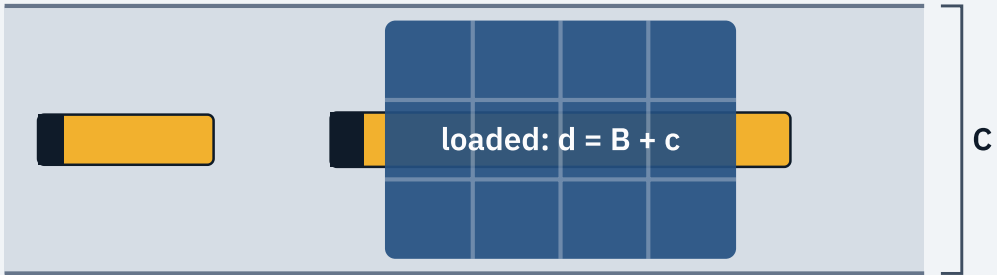
Harbin Engineering University · Nanyang Technological University

Bound checked case by case on 17,760 occupancy simulation points in two yards · thesis-core 83c16bf

One resource = one width container

At every moment: widths in use $\leq$ capacity

capacity $C = \text{road width } W + \text{lateral clearance } c$



empty: not enough width left, waits

This resource over one day ($H = 16 \text{ h}$)



$$\sum \ell \cdot u(d / C) \leq H$$

One row per resource: this is the LP of the network bound

This sub-study in one minute

1

The question

When loaded transporters hold segments and junctions exclusively, how many tasks a day can a yard's network complete? The bound should hold for a whole class of traffic rules and show where it binds.

2

The model

Segments, junctions and stops are width containers, held for a blocking time (passage plus clearance, plus the turn sweep at junctions); loaded and empty vehicles take different widths.

3

The result

T1': the occupancy-rate LP plus width bin-packing rows, crane rows and a window edge term. Okpo 1,997 $\rightarrow$ 1,720 tasks a day (14% tighter); none of 17,760 occupancy simulation points in two yards exceeds it.

4

Next

A system T1' without sampling noise and the route-free T1" (T23); the safe-release rule segment_safe, with no teleport (T22); the author rewrites the proof.

The bound is the upper edge of the capacity interval, and it names the road or the door where capacity binds.

The order of this briefing (click to jump)

1	Question and place in the thesis C1, the theorem list and where this sub-study sits	p. 4
2	The occupancy model Width containers, blocking times, asymmetric occupancy, traffic rules and conditions	p. 5
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The question: write road occupancy as a structural capacity constraint and compute the daily throughput the network can carry

Place in the thesis

Chapter 2: the occupancy model (resources, blocking times, traffic rules).
The bound section of chapter 3 (capacity): T1' and its validation, then T1'' and the system T1'.

C1 Two bounds and the capacity interval

$\text{Throughput} \leq \min\{\text{fleet bound } K \cdot H / \bar{c}, \text{network bound}\}$. The network bound is set by the blocking-time structure and can be computed by linear programming. Attainable capacity lies within the interval [best deadlock-free rule, network bound].

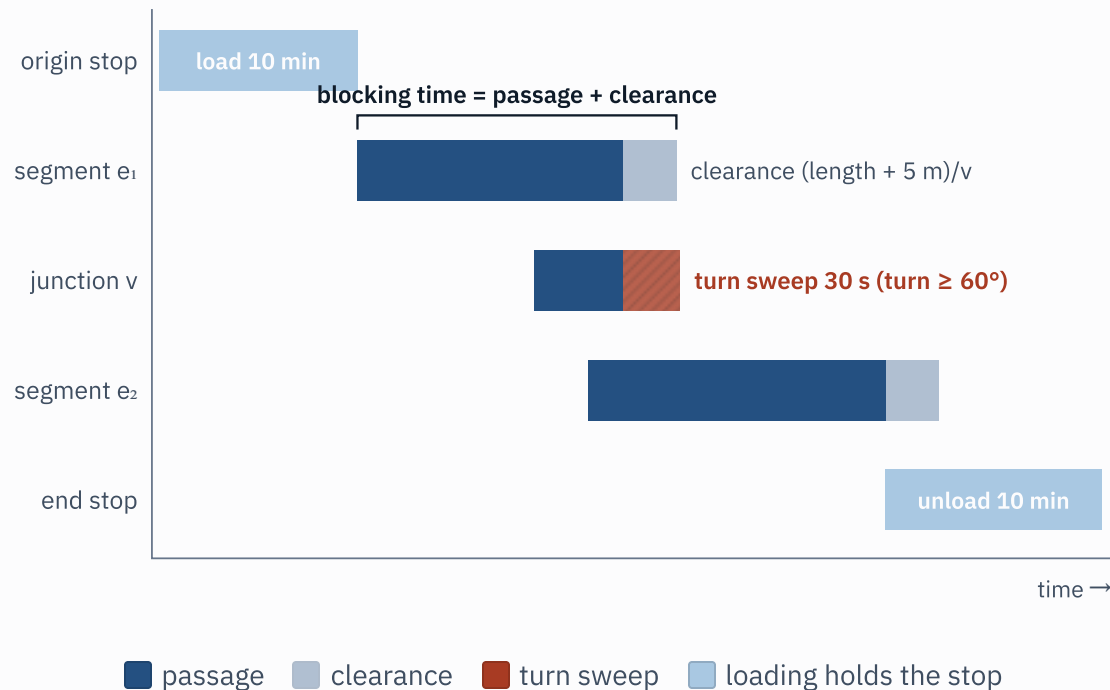
Also feeds

C4, the road–vehicle exchange rate: the bound's dual prices are its input. C5 and H6, cross-yard prediction: the network bound computed from the map, with its binding elements, is the predictor.

ITEM	STATEMENT (MANIFESTO V1, §4)	STATUS
T1'	Network part of the two bounds: occupancy-rate LP with width bin-packing rows, crane rows and the window edge term. Holds for rules satisfying (A1)–(A3); loaded routes fixed	draft; author rewrites
T1''	Route-free bound: loaded moves also as multicommodity flow; holds for any routing rule	to do
T6	Road–vehicle exchange rate: given by dual prices; upgrades pay only on tight constraints. Use T1'' for new links	author
Complexity	Feasibility with deadlines and exclusive resources is NP-hard (reduction from the blocking job shop)	to do
Candidate	Dock-mouth proposition: at nominal load $p < 1$, starvation caused by blocking destabilises the system; waiting off the road or using the dock road in sections restores stability. The other is attainability of T1'	after T22 item 2

Sources: research manifesto v1 (thesis-core tag manifesto-v1, docs/研究宣言.md), §2 C1 and the §4 theorem list, quoted in full; C4, C5 and H6 are named only with the input they need. Chapter plan from the direction document (8 Oct 2026), §4 “thesis architecture”.

Resources are width containers: segments, junctions and stops are held for a blocking time



- **Three kinds of resources:** segments (721 on Okpo), junctions and entrance nodes (552), drive-in stops (172); from T20 the cranes are dock resources too
- **Capacity** = road width W + lateral clearance c (1 m); a vehicle takes its outline width + c ; at every moment the widths in use stay within capacity
- **Blocking time:** a segment is held from the head entering until the tail clears the next junction, plus longitudinal clearance; a junction for $(D + \text{length} + 5 \text{ m})/v$, plus a 30 s sweep for turns $\geq 60^\circ$
- **Stops:** shops load at their doors (entrance nodes); stockyards and assembly areas have drive-in stops; buffers and docks load at the roadside, so that road segment is the stop

$$\sum_v d_{v,r}(t) \leq C_r \text{ for every resource } r \text{ and time } t$$

Loaded and empty occupancy are asymmetric: passability classes follow from widths, and widths and empty-run rules are scanned as scenarios



■ loaded: block width $B = 10$ m ■ empty: platform $w = 5.5$ m same scale; $c = 1$ m

- **Blocks** drawn with width $B \in \{5, \dots, 15\}$ m and length $\max(B, 10) - 30$ m; loaded outline width $\max(B, 5.5)$ m. Empty platform 16.25×5.5 m; 12 km/h empty, 6 km/h loaded
- **No precise survey:** road widths are graded from imagery (about ± 1.5 m); the three classes follow from widths, not from road-by-road judgement

SCENARIO	OKPO T1'	YANTAI T1'
Main scenario	1,720	1,252
Widths +1.5 m	1,789 +4%	1,252 ±0%
Widths -1.5 m	1,674 -3%	1,248 ±0%
Empties exclusive everywhere	1,720 ±0%	1,252 ±0%
Empties pass everywhere	1,822 +6%	1,252 ±0%

Yantai binds at door nodes, so neither widths nor the empty-run rule move it; Okpo binds on narrow roads and responds to both.

Sources: thesis-core 83c16bf, docs/T17_flag_report.md §1.1, §1.3 and the §2 parameter table (block sizes as in MY-B; speeds from the Suzhou Dafang DCY270 data sheet); T1' in the table is the mean of 10 seeds in results/a0ab16fceb94/bounds.csv (Okpo) and b82aa19b22af/bounds.csv (Yantai), tasks a day. Sketch drawn to one scale.

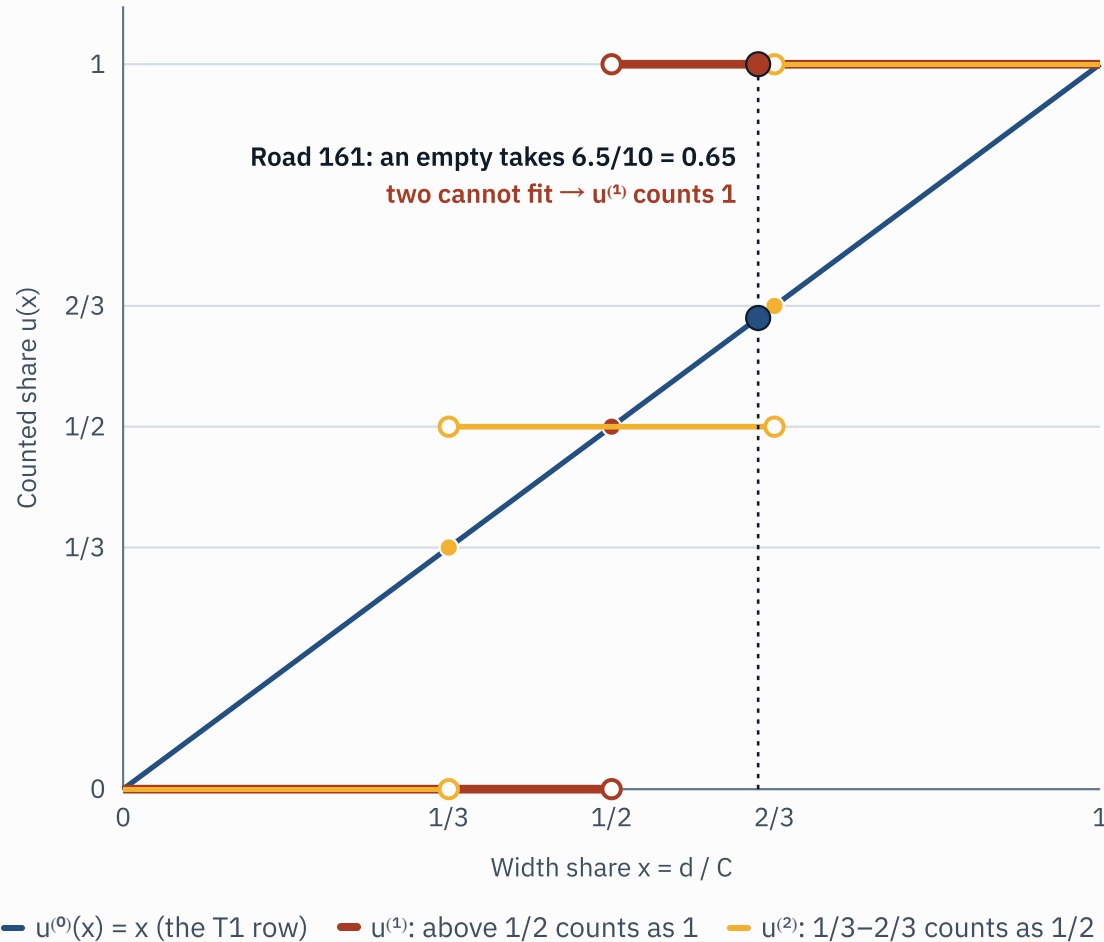
Four traffic rules, and the conditions (A1)–(A3) under which the bound holds

Free flow	No capacity check, vehicles never interact; the field's default. The bound does not apply: on a one-lane example it is exceeded already at $K = 4$	Conditions (proof draft, §3) (A1) Each task holds each resource at least for its no-wait window and at least at its outline width (A2) Empty runs occupy resources at least as much as the LP's empty-run coefficients (A3) The windows counted for one vehicle never overlap: the end window stops when unloading ends Whole-route reservation proved Segment request while no clearing occurs proved Reference rule (teleport clearing): a teleport can cut a hold short, so (A1) may fail checked per case segment_safe: no teleport, waiting only lengthens holds after T22 review Free flow: no capacity check does not apply
Whole-route reservation (just-in-time departure)	Empty run, loading, loaded run and unloading booked at once; waits only at stops, never on the way; deadlock-free by construction	
Segment request · reference rule (teleport clearing)	At each junction request “junction + next segment”, waiting on the current segment if refused; a deadlock sends one vehicle off the network for 900 s. Reference only	
segment_safe T22 in progress	Segment request with safe release (Banker-type deadlock avoidance): only requests that cannot lead to deadlock are granted, no teleport; plus a no-clearing switch	

All four share one dispatcher: nearest idle vehicle, first come first served. The fleet bound $K \cdot H / \bar{c}$ holds for all four.

Sources: thesis-core 83c16bf, docs/T18_T1_tight_proof.md §1, §3, §5; docs/T17_flag_report.md §1.2 and §0 (one-lane counterexample). segment_safe as defined in the T22 task statement (00_总控/runs/2026-10-09_T22_prompt.md §2.1); no reviewed results yet.

T1' : three new kinds of rows tighten the LP bound and bring in the dock



$$\begin{aligned}
 &\max \lambda \\
 &\text{s.t. } \lambda \cdot \bar{a}_r^{(k)} + \sum_a f_a \cdot e_{ar}^{(k)} \leq H \quad \forall r, k = 0, 1, 2 \\
 &\quad \sum_{a \rightarrow n} f_a - \sum_{n \rightarrow a} f_a = \lambda \cdot (o_n - d_n) \quad \forall n \\
 &\quad \lambda \cdot s_z \cdot 190 \min \leq m_z \cdot H \quad \forall \text{dock } z
 \end{aligned}$$

Width bin-packing rows

T18

The share $x = d/C$ becomes the Fekete–Schepers dual-feasible functions $u^{(1)}, u^{(2)}$, in exact rationals: two vehicles that cannot fit side by side each count as exclusive

Crane rows

T20

One row per dock: an erection holds a crane 190 min (hook 10, position and release 180) against cranes $\times 16$ h

Window edge term

T21

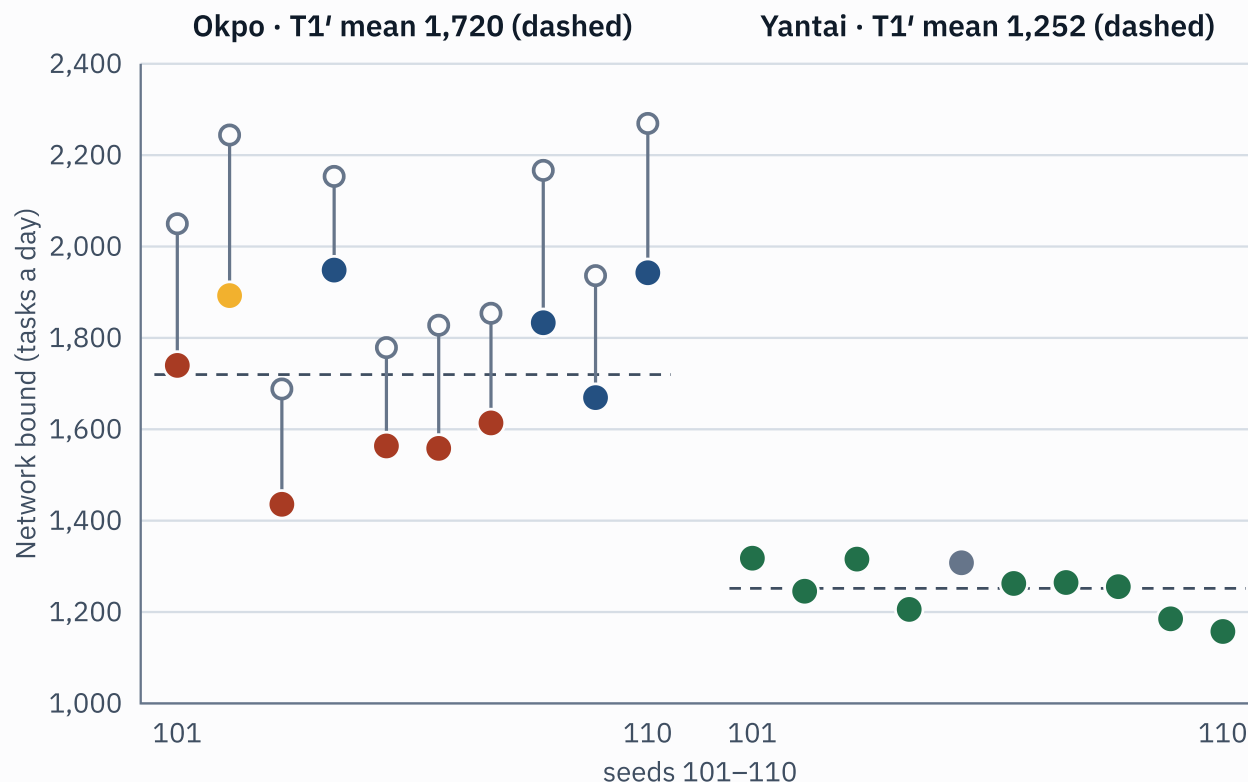
The case check runs over a finite window W , where a crane can start once more than its rate: each lift counts $190 \min \times W / (W + 190 \min)$

Theorem T1': for a rule satisfying (A1)–(A3), long-run daily throughput $\lambda \leq \min\{T1', K \cdot H / \bar{c}\}$; loaded routes as in the simulation (shortest paths).

Sources: thesis-core 83c16bf, docs/T18_T1_tight_proof.md §1–3 (lemma 1 is a special case of Fekete & Schepers 2001, proved on its own in the draft); crane rows in the T20 report §0 item 3, the 190 min from Zhang 2025, lines 2525–2528 (another yard); the window edge term in the T21 report §5. $\bar{a}$ and e are the no-wait holds of one task and one empty run on resource r (seconds $\times$ share u); s_z is the erection share to dock z and m_z its cranes.

The chart is drawn from the definitions.

Seed by seed in two yards: Okpo tightens by 14% and binds on roads; Yantai binds at the two doors of one building



1,997 → 1,720

Okpo main scenario, T1 → T1', mean of 10 seeds (a day, -14%); T1' ranges 1,436–1,948 across seeds

802 → 703

Okpo erection mix (P6 15%), -12%; 10 of 10 seeds bind on road 161

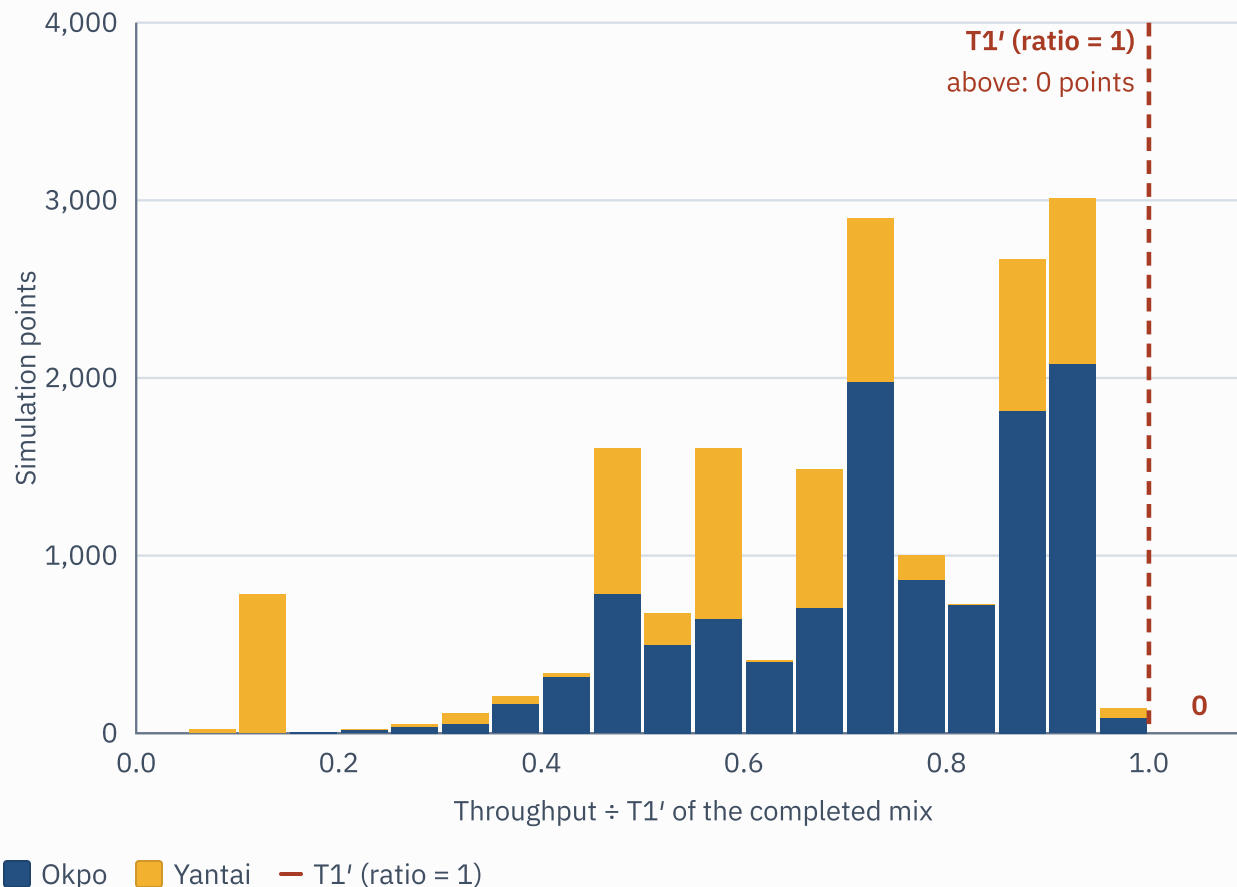
1,252

Yantai main scenario T1'; 9 of 10 seeds bind at the two doors of building 014. T1 = 1,246: T1 counts the clearance after unloading twice and serves only for comparison

○ T1 (hollow) ● T1' · road 161 ● T1' · road 010-6 ● T1' · road 026
● T1' · building 014 doors ● T1' · building 007 door

Sources: thesis-core 83c16bf, results/a0ab16fceb94/bounds.csv (Okpo, map rev 1824; T21 confirms rev 1825 has the same network), b82aa19b22af/bounds.csv (Yantai rev 395, same network as rev 396), f7ac88b60111/bounds.csv (erection mix). Conditions: MY-B flow mix, the first 2,000 tasks of each seed's stream; each road one resource. Yantai's binding nodes share one name in the results and were told apart by the project control rerunning the 83c16bf code (00_总控/tasks.md, 9 Oct 2026).

Checked case by case: none of 17,760 occupancy simulation points in two yards exceeds T1'



0 / 17,760

Points above T1', of 11,180 on Okpo and 6,580 on Yantai (reservation and the reference rule, peak and saturated)

0.973

Largest ratio: Okpo, $\rho = 0.99$, reference rule (teleport clearing), saturated $K = 20$; Yantai's largest 0.962

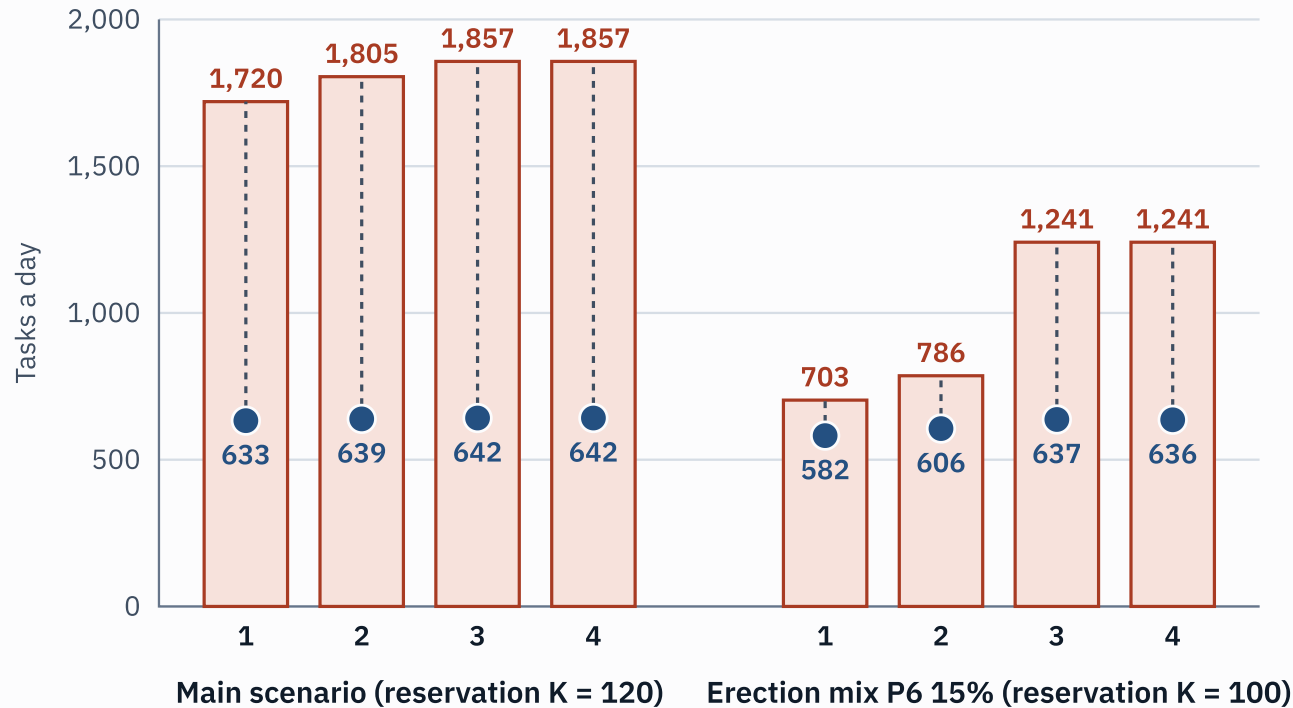
3,149 / 3,152

Of the points with ratio ≥ 0.9 , those from settings with nominal crane load $\rho \geq 0.92$: cranes nearly saturated, bound nearly reached

Before: T18's four batches, 12,140 points, largest 0.949; in T20, 13 points were over by up to 2.3%, traced to the window edge; with the edge term in the code (T21) none is over.

Sources: thesis-core 83c16bf, results/1b06c6d46f0d/points.csv (Okpo) and 87d02d32bc13/points.csv (Yantai): T21's crane-cadence batches, whole-route reservation and the reference rule (teleport clearing), peak and saturated demand; ratio = throughput ÷ T1' of the mix completed in that point's window (with crane rows and the edge term). Totals agree with T1_tight_check in t21_summary.json. Small instances in docs/T18_T1_tight_proof.md §4; the T18 and T20 checks in their reports, §2 and §8.

Three changes to road 161: the main-scenario bound moves only 5–8%; under the erection mix it depends on how the dock road is divided



- **Main scenario:** 1,720 → 1,805 (widened to 12 m) → 1,857 (sectioned at its stops) → 1,857 (plus two passing bays); binding moves to road 010-6. Reservation throughput moves only +0.2–2.4%: the rule sets the lower edge
- **Erection mix (P6 15%):** 703 → 786 → 1,241 (+77%) → 1,241: the bound depends on how the dock road is discretised; reservation 582 → 637 (+10%)
- **Still a bound after sectioning:** resources are only finer, (A1)–(A3) still hold, the proof is unchanged

A probe from the author's review: bounds over 10 seeds, simulations over 3, sections without direction locks. T25 gives the final numbers (10 seeds, full K grid).

■ T1' (10 seeds) ● Whole-route reservation, saturated throughput (3 seeds)
 — between them: the capacity interval **1** current model **2** widened to 12 m **3** sectioned at stops
4 sectioned + 2 passing bays

Source: the author's review (00_总控/audit/response/占路运输全量审核意见与后续指导_20261009.md), the two tables of appendix A, parsed by make_data.py; code thesis-core 83c16bf, Okpo map rev 1825. Reservation throughput: K = 120 in the main scenario and K = 100 in the erection mix, saturated, 4 days counting days 2–4; the +0.2–2.4% covers both K = 60 and 120. Widening: road 161 from 9 m to 12 m; sectioning: boundaries 25 m either side of its 3 stops, giving 7 resources; passing bays: a 60 m unrestricted piece in the middle of each of the two long gaps.

Both yards' binding elements sat on conservative modelling choices: the key-element review becomes a fixed step

Okpo · road 161 (9 m wide, 1,095 m, at dock no. 1)

Binds in 5 of 10 seeds in the main scenario, dual price 98.9 a day per extra hour a day, about 4.2 vehicles; road 010-6 in 4 of 10, 115.5. Erection mix: 10 of 10, 43.9, about 1.8 vehicles.

Author's imagery check: while a vehicle unloads or waits for the crane at the dock edge, the neighbouring assembly areas stay reachable. → From T22, Okpo's main scenario sections the road at its stops; the whole road is the comparison.

Yantai · the two doors of building 014

Binds in 9 of 10 seeds. Both doors are dead-end nodes where vehicles load and unload, holding the door about 10 min; the building has only two doors, each serving about 7% of all tasks.

Author's check on imagery: two doors, and vehicles can pass in front, so “exclusive loading at the door” is conservative. → T25 adds a door-granularity scenario.

The fixed step (every yard in chapter 6)

- 1 Compute T1' and the dual price of every resource
- 2 List the three elements with the highest dual prices
- 3 The author checks on imagery: **road width, number of entrances, how traffic passes a stop**
- 4 Choose the resource granularity, recompute, and record it in the map's source field

This is a pinpoint check of a few key elements, not a precise survey; it can go into the thesis as an engineering method (minimal human check).

What stands and what comes next: every computation lands in a figure, a table or a section of the thesis

In hand

- ✓ The T1' LP with its three kinds of rows, code and tests (exact dual-feasible functions, three hand-checked instances, the crane edge term)
- ✓ Proof draft for rules satisfying (A1)–(A3)
- ✓ 17,760 points in two yards checked, none over
- ✓ Okpo's main scenario is robust to widening, sectioning and passing bays on road 161 (+5–8%)

Conditions (stated once): loaded routes fixed to shortest paths; per-seed task mixes carry sampling noise (Okpo 1,436–1,948); the reference rule (teleport clearing) is outside the proof; Okpo's 1,720 treats each road as one resource (1,857 when sectioned); maps Okpo rev 1824/1825, Yantai rev 395/396; MY-B flow mix.

TASK	WHAT	OUTPUT (PLACE IN THE THESIS)	STATUS
T23	System T1' from expected holds, no sampling noise	Bound numbers in chapter 3	running
T23	Route-free T1''; value of route choice T1'' – T1'	Chapter 3 bound section; dual-price column of chapter 5's rate table	running
T23	Top-10 dual prices	The key-element review list	running
T22	segment_safe; the dock road sectioned at its stops	Lower edge of the capacity-interval figure (ch. 3); rule inside the proof	running
T25	Door granularity; unambiguous resource names	Yantai's T1' and binding elements	to do
Author	Rewrite the T1' proof	Chapter 3, theorem section	weeks 3–4
–	Candidate: attainability of T1' (a periodic rule reaching $1 - \varepsilon$ of T1' when the dock dominates)	Chapter 3 theorems; the case for TR-B / TS	after T22
–	Complexity: NP-hard (from the blocking job shop)	Chapter 3, theorem section	to do

What is new: an LP capacity bound for road networks with shareable widths and asymmetric loaded and empty occupancy

- **Within our search** (the author's review, appendix B, 9 Oct 2026), the shipyard transport literature has no capacity bound, and no direct counterpart of an LP throughput bound for AGV zone control turned up (that search was abstract-level)
- We extend the occupancy-rate LP to road networks with **shareable widths and asymmetric loaded and empty occupancy**, valid for any rule satisfying (A1)–(A3)
- Dual-feasible functions, made for bin-packing lower bounds, serve here as **time-integrated capacity rows**; crane rows and the window edge term bring the dock into the same bound
- **As a system**: tracing on public imagery → bound → dual prices → key-element review, one LP per yard

SOURCE WE BUILD ON	WHAT WE TAKE	READ
Railway blocking times and UIC 406 compression; Goverde 2007	Blocking time → occupancy rows; max-plus gives the cycle of a given order, a lower-side result	abstract; UIC 406 to check
Jensen, Schmidt, Nielsen 2020 (TR-C)	Railway network capacity without a fixed timetable: closest in structure	abstract
Campos, Chiola, Silva	LP throughput bounds for Petri nets: closest in structure	garbled PDF; venue to check
Queueing networks with blocking: Onvural 1990; Balsamo et al. 2001	Theory of blocking and starvation	abstract; book to check
Fekete & Schepers 2001	Dual-feasible functions	lemma proved in the draft
Mascis & Pacciarelli 2002	Blocking job shop → complexity reduction	abstract

Candidate journals for a bound-and-model paper: several options, chosen when the paper is written

KIND	JOURNAL	WHEN IT FITS
First · theory	Transportation Research Part B	The capacity bound and interval as the main line; stronger once T1' attainability or the dock-mouth proposition is proved
First · theory	Transportation Science	Network capacity, traffic rules and the bound side by side; same conditions
Alternative · OR	Operations Research	If T1', T1'' and attainability are all proved, as a general LP capacity method
Alternative · OR	Naval Research Logistics	The method: dual-feasible-function rows, dual prices and the bound
Alternative · OR	European Journal of Operational Research	Model, bound and computation on two yards in balance
Applied	IISE Transactions	Capacity analysis and upgrade decisions for manufacturing logistics (key elements, dual prices)

The author's guidance (§4.2) for the first paper, which combines model, capacity interval, dock and exchange rate: submit to TR-E or IISE Transactions; with attainability of T1' and the dock-mouth proposition added, aim at TR-B or Transportation Science. This slide lists options only and fixes no target.

Sources: the author's review §4.2 (venues for the first paper); the direction document §4 (chapter 2 published with chapter 3, whose first choice is TR-B or TS); “when it fits” is our own matching of this sub-study to each journal, to be checked against each journal's aims and scope before submission.

References and data sources

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Data and documents: thesis-core commit 83c16bf (T21, reviewed 9 Oct 2026): results/a0ab16fceb94, b82aa19b22af, f7ac88b60111 (bounds.csv), 1b06c6d46f0d, 87d02d32bc13 (points.csv), t18_report_d10b9015cd4d, t21_report_9eb8bf099d3a (summaries); docs/T17_flag_report.md, T18_report.md, T18_T1_tight_proof.md, T20_report.md, T21_report.md; manifesto v1 (tag manifesto-v1); the author's review of 9 Oct 2026 (appendices A and B); 00_总控/tasks.md; Suzhou Dafang DCY270 data sheet (speeds). Chart numbers are generated by bound/make_data.py.

What this sub-study gives the thesis: a network bound that can be computed, checked and that points the way

1**Computable**

One LP per yard, using only networks traced on public imagery and public parameters.

2**Checked**

The proof draft covers rules satisfying (A1)–(A3); none of 17,760 occupancy simulation points in two yards exceeds it.

3**Points the way**

Dual prices name the binding roads and doors, leading to the key-element review and the road–vehicle exchange rate.

[← Back to the thesis briefing](#)[The capacity-interval sub-study →](#)[The fleet–network co-design sub-study \(exchange rate\) →](#)

Transporters own the road: capacity is capped by the network, and the bound tells us how high the cap is and where it binds.

Thank you.